Engine Live Data Demonstration

Student learning guide | ICE laboratory | Local interface review

From your first sample to interpreting controller commands. Prepared 30 October 2026. This guide accompanies the section-based input/output workspace.

Start with the two sides

Left: change sensor inputs and starting switches. Right: choose a section in the ribbon. That section shows its process display and matching graphs. All values come from one shared model; changing tabs does not start another engine or change the calculations.

Your first run

Load a preset, Run simulation, switch ignition and simulated authorisation on, then press Start engine. Defaults impose 800 rpm; after priming and cranking the sequence can reach RUN. RPM is a student input: the model does not calculate mechanical acceleration.

What does Step mean?

This module uses a fixed calculation interval of 0.05 seconds: 20 samples per simulated second. Step records exactly one sample while paused. The first sample is at 0 s; subsequent samples are 0.05 s apart (21 samples span 0 to 1 s). It does not mean a sudden coolant-temperature change. Time in state means how long the current OFF, PRIME, CRANK, RUN or SHUTDOWN state has lasted. Inputs are queued and applied at the next sample.

Fuel and injection

Use the Fuel and injection ribbon tab. Injector pulse width means how long the injector is commanded open for each event, in milliseconds. Positive STFT adds fuel; negative STFT removes fuel when closed loop is active. Try O2 voltage 0.10 V after RUN and compare with the warm default 0.45 V. The illustration is a command diagram, not a measured spray or fuel-consumption model.

Idle and ignition

Compare target RPM with imposed actual RPM. Error is target minus actual RPM while idle control is active. Accessory load can increase the duty command. A larger duty does not automatically accelerate the imposed RPM input. Ignition timing is degrees before top dead centre; negative timing means after it.

Cooling

Virtual ECT sensor voltages become two temperatures. The hotter value controls cooling. Fan and virtual electric-pump percentages are commands, not measured speeds or flow. A difference above 15 °C flags disagreement. The separate physical 1NZ rig has no electric water pump.

Start and run

OFF waits for a permitted start request. PRIME lasts 2 s. CRANK permits the starter and times out at 5 s if imposed RPM does not reach the running threshold. SHUTDOWN lasts 0.5 s. Start requires ignition, simulated authorisation, at least 9.5 V and no emergency. Emergency stop is latched until deliberately cleared. These are virtual switch inputs, not voice or biometric recognition.

Read the graphs and legends

Each panel uses its own labelled unit and vertical scale. Solid, dashed and dotted lines distinguish signals as well as colour. Hover or focus a legend label for its meaning. Move the sample slider to inspect history; live simulation continues unless paused. History retains the latest 60 s, up to 1201 samples. CSV exports retained sampled history rather than a hardware log.

Verify one change at a time

Pause to inspect the current result. Load a preset for a fresh experiment; a latched emergency remains set. Reset restores all defaults and clears history. Try low battery to see start inhibition; change O2 voltage to see fuel correction; change one ECT voltage to inspect disagreement. Compare expected versus observed behaviour and keep your input configuration with exports.

Advanced scope and evidence

The pure reference controller preserves input caps, equations, state transitions and fixed 0.05 s sampling. Numerical and browser tests verify the educational software; they do not prove OEM calibration, engine combustion, hardware response or real-vehicle safety. Process animation is qualitative and stops with pause or reduced motion. The calculations are unchanged in this interface revision.

Complete input and output glossary

Input limits and defaults are taken directly from the preserved module specification. Outputs are calculated or commanded values, rather than measured hardware behaviour.

Battery voltage

Used for safe-start permission and injector battery compensation.

Signal: BatteryVoltage | Unit: V

Default: 13.8 | Accepted range: 8 to 14.9

Engine speed

Student-selected RPM; affects starting, injection, idle and ignition.

Signal: EngineSpeed | Unit: rpm

Default: 800 | Accepted range: 0 to 6500

Throttle position

TPS: throttle position; affects load, injection and idle activation.

Signal: ThrottlePosition | Unit: %

Default: 2 | Accepted range: 0 to 100

Manifold pressure

MAP: manifold absolute pressure; affects load, injection and timing.

Signal: ManifoldPressure | Unit: kPa

Default: 35 | Accepted range: 20 to 105

Oxygen-sensor voltage

Simplified oxygen feedback used to estimate lambda and fuel correction.

Signal: O2SensorVoltage | Unit: V

Default: 0.45 | Accepted range: 0.05 to 0.95

Virtual ECT1 sensor voltage

Virtual ECT1 input; affects fuel, idle and cooling calculations.

Signal: ECT1Voltage | Unit: V

Default: 1 | Accepted range: 0.2 to 4.8

Virtual ECT2 sensor voltage

Virtual ECT2 input; used for cooling demand and sensor plausibility.

Signal: ECT2Voltage | Unit: V

Default: 1.1 | Accepted range: 0.2 to 4.8

Accessory/electrical load

Represents the added idle-control demand of electrical accessories.

Signal: AccessoryLoad | Unit: %

Default: 10 | Accepted range: 0 to 100

Ignition switch

Simulated ignition status; turning it off requests shutdown.

Signal: IgnitionSwitch | Unit: state

Default: 0 | Accepted range: 0 to 1

Simulated authorisation

Simulated permission; no biometric recognition is performed.

Signal: StartAuthorisation | Unit: state

Default: 0 | Accepted range: 0 to 1

Virtual start request

A ten-sample virtual request; it cannot bypass safe-start checks.

Signal: StartRequest | Unit: state

Default: 0 | Accepted range: 0 to 1

Normal stop request

A 0.5-second virtual request for normal shutdown.

Signal: StopRequest | Unit: state

Default: 0 | Accepted range: 0 to 1

Emergency stop

Latched emergency request overrides engine and cooling commands.

Signal: EmergencyStop | Unit: state

Default: 0 | Accepted range: 0 to 1

Air-conditioning request

Requests at least 35% fan duty when state code is CRANK or higher.

Signal: ACRequest | Unit: state

Default: 0 | Accepted range: 0 to 1

Engine state

OFF, PRIME, CRANK, RUN or SHUTDOWN.

Signal: EngineState | Unit: code

Elapsed time in state

Virtual seconds since the current engine state was entered.

Signal: StateElapsedTime | Unit: s

Fuel-pump command

Fuel-pump relay is on during PRIME, CRANK and RUN.

Signal: FuelPumpState | Unit: state

Starter command

Starter relay is on only during CRANK.

Signal: StarterState | Unit: state

Injection enabled

Injection is allowed only during CRANK and RUN.

Signal: InjectionEnable | Unit: state

Ignition enabled

Ignition is allowed only during CRANK and RUN.

Signal: IgnitionEnable | Unit: state

Sequence active

Indicates an active PRIME, CRANK or RUN sequence.

Signal: StartPermitted | Unit: state

Engine load

Weighted load estimate from MAP and TPS.

Signal: EngineLoad | Unit: %

Lambda estimate

Simplified air/fuel indication: above 1 is lean, below 1 is rich.

Signal: LambdaEstimate | Unit: ratio

STFT / equivalent closed-loop fuel correction

Positive correction adds fuel; negative correction removes fuel when closed loop is active.

Signal: STFT | Unit: %

Base injector pulse width

Uncorrected injection duration derived from RPM, MAP and TPS.

Signal: BaseInjectorPW | Unit: ms

Warm-up enrichment

Additional fuel percentage below 70 °C.

Signal: WarmupEnrichment | Unit: %

Battery compensation

Injector delay compensation at lower battery voltage.

Signal: BatteryCompensation | Unit: ms

Final injector pulse width

Corrected injection duration; zero when injection is disabled.

Signal: InjectorPW | Unit: ms

Closed-loop active

Requires enabled injection, ECT1 at least 70 °C and RPM at least 600.

Signal: ClosedLoopActive | Unit: state

Target idle speed

Coolant-dependent idle target, from 750 to 1200 rpm.

Signal: TargetIdleRPM | Unit: rpm

RPM error

Target minus actual RPM while idle control is active.

Signal: RPMError | Unit: rpm

Idle-control duty

Idle actuator duty based on speed error and accessory load.

Signal: IdleControlDuty | Unit: %

Ignition timing

Commanded degrees BTDC; a negative value represents ATDC.

Signal: IgnitionTiming | Unit: deg BTDC

Idle active

Requires RUN, TPS at most 5% and RPM below 2000.

Signal: IdleActive | Unit: state

Virtual ECT1 temperature

Educational temperature calculated from virtual ECT1 voltage.

Signal: ECT1Temp | Unit: degC

Virtual ECT2 temperature

Educational temperature calculated from virtual ECT2 voltage.

Signal: ECT2Temp | Unit: degC

ECT sensor difference

Absolute difference between the two calculated temperatures.

Signal: SensorDifference | Unit: degC

ECT sensor disagreement

Set when virtual ECT temperatures differ by more than 15 °C.

Signal: SensorDisagreement | Unit: state

Control temperature

The hotter calculated coolant temperature controls cooling.

Signal: ControlTemperature | Unit: degC

PWM fan duty

Virtual pulse-width-modulated fan command.

Signal: FanDuty | Unit: %

Virtual electric-water-pump duty

Virtual electric pump command; no electric pump exists in the physical 1NZ setup.

Signal: WaterPumpDuty | Unit: %

Cooling-loop active

At least one virtual cooling actuator has nonzero duty.

Signal: CoolingLoopActive | Unit: state

Additional derived signals

Imposed actual RPM (ActualRPM): mirrors the student-selected speed. Safe start condition (SafeStartCondition): includes the momentary start request and returns false after that request even during RUN.

Reference evidence and verification

Preserved engineering metadata: 2026-10-12-existing-platform-section. The module specification, pure reference equations, model and controller are the software evidence for the display. No new OEM or vehicle-specific calibration is claimed.

A repeatable check: load warm defaults; Run; enable ignition and simulated authorisation; Start. At RUN, final injector pulse width is about 1.679 ms. Pause, change O2 voltage from 0.45 V to 0.10 V and Step once. Final pulse width becomes about 1.926 ms and STFT is about +14.7%. Restore the preset before the next independent experiment.

Use a low-battery preset to verify inhibited starting. Use a sensor-disagreement preset to inspect both temperatures, their difference and full cooling commands. Export retained CSV samples for your report and record your inputs, selected preset, observed states and any deviations. Emergency stop remains latched across preset loads until explicitly cleared.

The interface revision changes presentation and guidance only. Software tests establish the educational implementation behaviour; they are not experimental validation against an OEM controller or a physical vehicle.